

Reg no : _____

Name : _____

AMAL JYOTHI COLLEGE OF ENGINEERING, KANJIRAPPALLY

IOT PROJECT - COMBINED ASSIGNMENT

24MCAT213 - INTERNET OF THINGS

MCA2025-27-S3

Max Marks: 6

Answer all questions.

Q.No	Question	Mark
1.	Develop an IoT project using an ESP32 and at least one sensor module to collect and transmit sensor data. Configure a containerized backend with database support to store and manage the collected data, and implement a real-time dashboard for visualizing the sensor readings. Your implementation should include the following: 1. ESP32 and Sensor Module – Interface at least one sensor with ESP32 and acquire sensor readings. 2. Data Communication – Transmit the sensor data from ESP32 to the backend using a suitable communication protocol. 3. Containerized Backend – Deploy the backend application using Docker or another containerization platform. 4. Database Support – Store the sensor data in a suitable database. 5. Real-Time Dashboard – Develop a dashboard that displays sensor readings in real time using suitable charts, graphs, or indicators. 6. Testing and Demonstration – Demonstrate the complete data flow from the sensor to the ESP32, backend, database, and dashboard. 7. Documentation – Submit a project report documenting the system architecture, hardware/software requirements, implementation methodology, database design, screenshots, testing results, and observations.	(6)

Expected Output:

A working IoT prototype with ESP32, sensor module, containerized backend, database, and real-time sensor-data visualization dashboard, along with a brief project report documenting the architecture, implementation, and results.

COs : CO1,CO2,CO3,CO4,CO5, BLs : Creating